

Indian Statistical Institute, Bangalore Centre.
Back-paper Exam : Stochastic Processes (B3)

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Date : June 07, 2024.

Max. points : 50.

Time Limit : 2.5 hours.

Answer all questions. All questions carry 10 points.

Give necessary justifications and explanations for all your arguments. If you are citing results from the class, mention it clearly. Always define the underlying random variables and events clearly before computing anything ! Some common definitions and inequalities are listed at the end.

1. (a) For a random variable X , define $\|X\|_E := \inf\{t > 0 : \mathbb{E}[e^{|X|/t}] \leq 2\}$. Show that $\|\cdot\|_E$ defines a norm on the space of sub-Exponential random variables. (5)
- (b) Let $A = (a_{i,j})$ be a symmetric matrix. Assume that for all $x_i \in \{-1, +1\}$, $|\sum_{i,j} a_{i,j} x_i x_j| \leq 1$. Show that there exists an universal finite constant K (not depending on A) such that for all $N \geq 1$, and $u_i, v_j \in \mathbb{R}^N$ with $\|u_i\| = \|v_j\| = 1$, it holds that

$$|\sum_{i,j} a_{i,j} \langle u_i, v_j \rangle| \leq 2K. \quad (5)$$

2. Let $\mathbf{X} = [0, 1]^2$ and define the *toroidal metric* on $\mathbf{X}$ as follows : $d(x, y) = \min\{|x - y + z| : z \in \mathbb{Z}^2\}$. Let $X_1, \dots, X_n$ be i.i.d. uniform points in $\mathbf{X}$. Consider the graph whose vertex set is $1, \dots, n$ and edge between i and j if $|X_i - X_j| \leq r_n$ where $r_n \in (0, \infty)$ is such that $(n-1)\pi r_n^2 = \lambda \in (0, \infty)$. Let I_n be the number of isolated vertices. Show that

$$n^{-1} I_n \rightarrow e^{-\lambda}, \text{ a.s..}$$

3. Let $V = \mathbb{N} \cup \{0\}$, $p_i, q_i > 0$, $r_i \geq 0$ and $p_i + q_i + r_i = 1$ for all $i \in V$. X_t be the total population at time t which evolves as follows : If $X_t = i$, then at time $t+1$, there is a birth (i.e., population increases by 1) with probability p_i , death with probability q_i (i.e., population decreases by 1) and nothing happens with probability r_i (i.e., population does not change). Show that X_t is irreducible. Determine the parameters for which X_t has a stationary distribution and determine the stationary distribution. Also determine when is X_t reversible.
4. Let X_t be the birth-death chain on $\mathbb{Z}_+$ with $P(x, x+1) = p, P(x, x-1) = 1-p$ and $P(0, 1) = 1$ for $p \in (0, 1)$. Let $h(0) = 1$.
 - (a) When $p > 1/2$, show that there is more than one bounded extension of h to $\mathbb{N}$ that is harmonic on $\mathbb{N}$.
 - (b) When $p \leq 1/2$, show that there is a unique bounded extension of h to $\mathbb{N}$ that is harmonic on $\mathbb{N}$.
5. Consider Polya's urn starting with 1 red ball and 1 green ball. At each step, a ball is chosen uniformly at random and the ball is returned to the urn along with another ball of the same colour. Let G_t be the number of green balls at time t , $t \in \mathbb{Z}_+$. Let $M_t = \frac{G_t}{t+2}$. Show that M_t is a Martingale and it converges in distribution to a uniform random variable in $[0, 1]$. Does M_t converge a.s. ?

SOME USEFUL NOTIONS

Sub-Gaussian random variable with variance factor ν : $\Psi_{X-\mu}(s) \leq s^2 \nu / 2$, for $s \in \mathbb{R}$ and where Ψ is the cumulant generating function i.e., $\Psi_X(s) = \log \mathbb{E}[e^{sX}]$.

Sub-exponential random variable with parameters ν and α : $\Psi_{X-\mu}(s) \leq s^2\nu/2$, for $|s| \leq \alpha^{-1}$. Equivalently, X is sub-exponential if there exists K such that $\Psi_{|X|}(s) \leq Ks$ for $0 \leq s \leq K^{-1}$.

Space of sub-Exponential random variables: This is set $\{X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \mathbb{R} : X \text{ is sub-Exponential}\}$ with the equivalence relation $X \sim Y$ if $X = Y$ a.s. and $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space.

Hölder's inequality: Let $w_j \geq 0$, $\sum_{j=1}^n w_j \leq 1$ be weights and Y_j 's be non-negative random variables such that $\mathbb{E}[Y_j^{1/w_j}] < \infty$ for all j . Then, we have that

$$\mathbb{E}\left[\prod_{j=1}^n Y_j\right] \leq \prod_{j=1}^n \mathbb{E}[Y_j^{1/w_j}]^{w_j}.$$

Gamma function: $\Gamma(a) = \int_0^\infty e^{-s} s^{a-1} ds$ for $a > 0$ and $\Gamma(a) \leq a^a$.

Stirling's Approximation: $\sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{1/(12n+1)} \leq n! \leq \sqrt{2\pi n} \left(\frac{n}{e}\right)^n e^{1/(12n)}.$

Laplacian: For $f : V \rightarrow \mathbb{R}$ and stochastic matrix P , define the Laplacian operator as $\Delta f(x) := \sum_y P(x, y)f(y) - f(x)$.

Harmonic function: A function $f : V \rightarrow \mathbb{R}$ is *harmonic* at x if $\Delta f(x) = 0$. It is harmonic on a set $B \subset V$ if it is harmonic at x for all $x \in B$.